

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

SRINIVASAN R

25ITR141

PROJECT REPORT

Submitted by

JAVA ASSIGNMENT

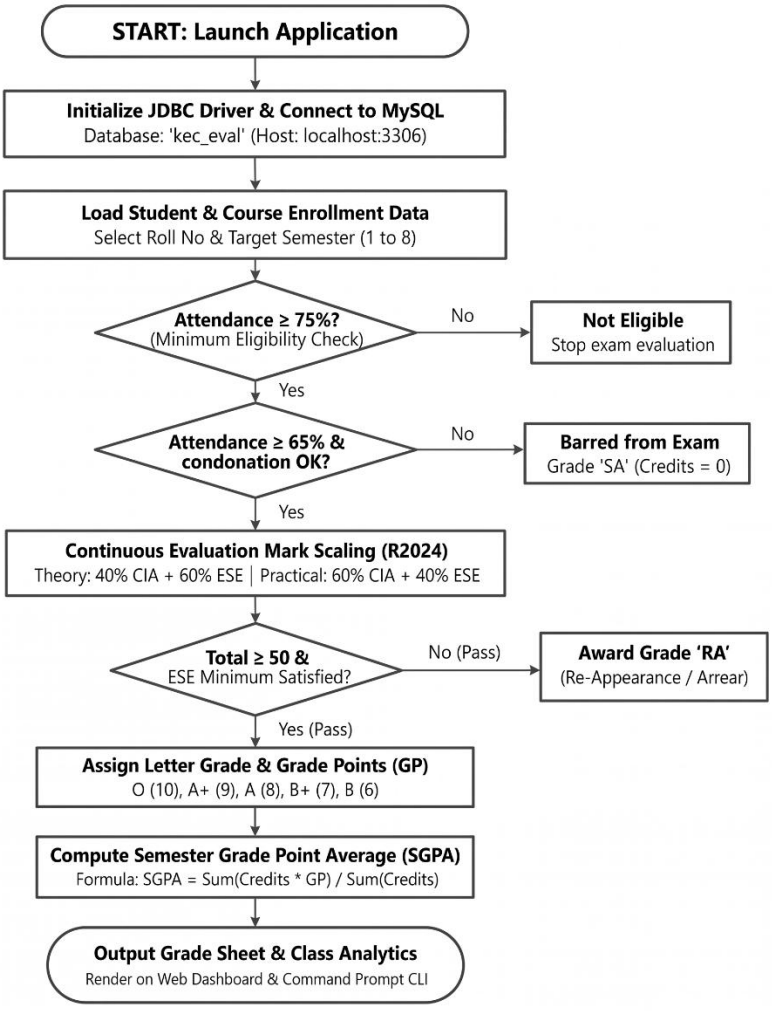
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
mysql> describe students;
+------------------+-------------+------+-----+-------------------+-------------------+
| Field            | Type        | Null | Key | Default           | Extra             |
+------------------+-------------+------+-----+-------------------+-------------------+
| roll_no          | varchar(20) | NO   | PRI | NULL              |                   |
| name             | varchar(100)| NO   |     | NULL              |                   |
| department       | varchar(20) | NO   |     | NULL              |                   |
| current_semester | int         | NO   |     | 3                 |                   |
| created_at       | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.03 sec)
```

2. COURSES TABLE:

```
mysql> describe courses;
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                              | Null | Key | Default           | Extra             |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| course_code | varchar(20)                       | NO   | PRI | NULL              |                   |
| title       | varchar(150)                      | NO   |     | NULL              |                   |
| credits     | int                               | NO   |     | NULL              |                   |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED') | NO   |     | NULL              |                   |
| semester    | int                               | NO   |     | 3                 |                   |
| created_at  | timestamp                         | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.01 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+------------------+-------------+------+-----+-------------------+-------------------+
| Field            | Type        | Null | Key | Default           | Extra             |
+------------------+-------------+------+-----+-------------------+-------------------+
| enrollment_id    | int         | NO   | PRI | NULL              | auto_increment    |
| roll_no          | varchar(20) | NO   | MUL | NULL              |                   |
| course_code      | varchar(20) | NO   | MUL | NULL              |                   |
| semester         | int         | NO   |     | 3                 |                   |
| total_classes    | int         | NO   |     | 0                 |                   |
| attended_classes | int         | NO   |     | 0                 |                   |
| is_condoned      | tinyint(1)  | NO   |     | 0                 |                   |
| created_at       | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-------------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
| Field           | Type        | Null | Key | Default         | Extra                                   |
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
| mark_id         | int         | NO   | PRI | NULL            | auto_increment                          |
| enrollment_id   | int         | NO   | UNI | NULL            |                                         |
| cia1_marks      | decimal(5,2) | YES  |     | NULL            |                                         |
| cia2_marks      | decimal(5,2) | YES  |     | NULL            |                                         |
| assignment_marks | decimal(5,2) | YES  |     | NULL            |                                         |
| ese_marks       | decimal(5,2) | YES  |     | NULL            |                                         |
| is_ese_absent   | tinyint(1)  | NO   |     | 0               |                                         |
| updated_at      | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
8 rows in set (0.02 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.util.*;

import java.text.SimpleDateFormat;

/**

* KEC (Autonomous) - Regulation 2024

* Student Academic & Continuous Evaluation System

* Short documentation version of MainApp.java

*/

public class MainApp {

private static final AcademicDAO dao = new AcademicDAO();

private static final Scanner sc = new Scanner(System.in);

public static void main(String[] args) {

if (!DBConnection.testConnection()) {

System.out.println("Database connection failed. Check DBConnection.java.");

return;

}

System.out.println("KEC STUDENT EVALUATION SYSTEM - R2024");

boolean running = true;
```

```
while (running) {
System.out.println("\\n1.Student 2.Course 3.Enrollment/Attendance");
System.out.println("4.Enter Marks 5.Mark Sheet 6.Defaulters");
System.out.println("7.Department Summary 0.Exit");
System.out.print("Choice: ");
switch (sc.nextLine().trim()) {
case "1": registerStudent(); break;
case "2": addCourse(); break;
case "3": enrollStudent(); break;
case "4": enterMarks(); break;
case "5":
System.out.print("Roll number: ");
String roll = sc.nextLine().trim();
System.out.print("Semester: ");
printStudentMarkSheet(roll, Integer.parseInt(sc.nextLine().trim()));
break;
case "6":
System.out.print("Semester: ");
printAttendanceDefaulters(Integer.parseInt(sc.nextLine().trim()));
break;
case "7":
System.out.print("Semester: ");
printDepartmentSummary(Integer.parseInt(sc.nextLine().trim()));
break;
case "0": running = false; break;
default: System.out.println("Invalid choice.");
}
}
```

```
System.out.println("System closed.");
}
private static void enterMarks() {
System.out.print("Roll number: "); String roll = sc.nextLine().trim();
System.out.print("Course code: "); String course = sc.nextLine().trim();
System.out.print("Semester: "); int sem = Integer.parseInt(sc.nextLine().trim());
int id = dao.getEnrollmentId(roll, course, sem);
if (id <= 0) { System.out.println("Enrollment not found."); return; }
System.out.println("1.CIA marks 2.Assignment marks 3.ESE marks");
String phase = sc.nextLine().trim();
if (phase.equals("1")) {
System.out.print("CIA-1: "); double c1 = Double.parseDouble(sc.nextLine().trim());
System.out.print("CIA-2: "); double c2 = Double.parseDouble(sc.nextLine().trim());
System.out.println(dao.recordPhase1Marks(id, c1, c2) ? "Saved." : "Failed.");
} else if (phase.equals("2")) {
System.out.print("Assignment: "); double a = Double.parseDouble(sc.nextLine().trim());
System.out.println(dao.recordPhase2Marks(id, a) ? "Saved." : "Failed.");
} else if (phase.equals("3")) {
System.out.print("Absent in ESE? (y/n): "); boolean absent =
sc.nextLine().trim().equalsIgnoreCase("y");
Double marks = null;
if (!absent) {
System.out.print("ESE marks: "); marks = Double.parseDouble(sc.nextLine().trim());
}
System.out.println(dao.recordPhase3Marks(id, marks, absent) ? "Saved." : "Failed.");
} else System.out.println("Invalid phase.");
}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;


/**

 * DBConnection - Database Connection Management for KEC Evaluation System

 * Kongu Engineering College (Autonomous) - Regulation 2024

 */

public class DBConnection {

    private static final String DEFAULT_HOST = "localhost";

    private static final String DEFAULT_PORT = "3306";

    private static final String DEFAULT_DB = "kec_eval";

    private static final String DEFAULT_USER = "root";

    private static final String DEFAULT_PASS = "0000";


    public static String getUrl() {

        String host = System.getenv("DB_HOST") != null ? System.getenv("DB_HOST") : DEFAULT_HOST;

        String port = System.getenv("DB_PORT") != null ? System.getenv("DB_PORT") : DEFAULT_PORT;

        String db = System.getenv("DB_NAME") != null ? System.getenv("DB_NAME") : DEFAULT_DB;

                    return "jdbc:mysql://" + host + ":" + port + "/" + db +
"?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";

    }


    public static String getUser() {

        return System.getenv("DB_USER") != null ? System.getenv("DB_USER") : DEFAULT_USER;

    }
```

```
    public static String getPassword() {
            return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") :
DEFAULT_PASS;
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found in classpath: " + e.getMessage());
        }
    }

    /**
     * Obtains a new database connection
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(getUrl(), getUser(), getPassword());
    }

    /**
     * Tests the database connectivity
     */
    public static boolean testConnection() {
        try (Connection conn = getConnection()) {
            return conn != null && !conn.isClosed();
        }
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
-------------------------------- MAIN MENU ----------------------------------------
 1. Student Registration (Add New IT Student)
 2. Course Catalog Management (Semester 3 Courses)
 3. Course Enrollment & Attendance Tracking
 4. Incremental Marks Entry (Phase 1: CIA1/2 | Phase 2: Asg/Lab | Phase 3: ESE)
 5. Generate Student Semester Mark Sheet (SGPA & Equivalent %)
 6. Generate Attendance Defaulter & Condonation Report
 7. Generate Class Performance & Result Summary
 8. Run Full Automated Demonstration (Proof of Execution)
 9. Launch Interactive Web Dashboard (Browser GUI @ http://localhost:8080)
 0. Exit
-------------------------------------------------------------------------------
Enter your choice (1-8): 1
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
--- [1] Student Registration ---
Enter Roll Number (e.g., 24ITR050): 25ITR141
Enter Student Full Name: Lingesh B
Enter Department [Press Enter for IT]:
Enter Current Semester [Press Enter for 3]:
[SUCCESS] S25ITR141  ITL187 (Lingesh B) registered successfully.

Press [Enter] to return to the Main Menu...
```

2. COURSE CATALOG:

```
--- [2] Course Catalog Management ---
Enter Course Code (e.g., 24ITR31 / 24ITT32): 24ITT11
Enter Course Title: Game Development
Enter Credits (e.g., 2, 3, 4): 2
Enter Course Type (1: THEORY, 2: PRACTICAL, 3: EMBEDDED): 2
Enter Offering Semester (1-8): 4
[SUCCESS] Course 24ITT11 [PRACTICAL, 2 Credits] added successfully.

Press [Enter] to return to the Main Menu...
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
--- [3] Course Enrollment & Attendance Tracking ---
Enter Student Roll Number: 25ITR141
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Total Classes Held: 60
Enter Classes Attended: 52
Calculated Attendance: 86.67%
[INFO] Attendance >= 75%. Eligible for ESE.
[SUCCESS] Enrollment and Attendance updated. (Enrollment ID: 413)

Press [Enter] to return to the Main Menu...
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 1
Enter Student Roll Number: 25ITR141
Enter Course Code: 24ITC31
Enter Semester: 3
Enter CIA Test 1 Marks (0-100): 92
Enter CIA Test 2 Marks (0-100): 93
[SUCCESS] Phase 1 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 2
Enter Student Roll Number: 25ITR141
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Assignment / Quiz / Lab Evaluation Marks (0-100): 90
[SUCCESS] Phase 2 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

3. PHASE 3 (POST SEMESTER):

```
---------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) ----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 3
Enter Student Roll Number: 25ITR141
Enter Course Code: 24ITC31
Enter Semester: 3
Is student Absent (AB) in ESE? (y/n): n
Enter End Semester Exam (ESE) Marks (0-100): 87
[SUCCESS] Phase 3 ESE Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
Enter your choice (1-8): 6

Enter Semester to check (e.g., 3): 3

=============================================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                         ATTENDANCE DEFAULTER & CONDONATION REPORT
                                  SEMESTER: 3
=============================================================================================================
Roll No      Student Name         Dept     Course     Held     Attended Attd %     Condonation  Eligibility / Grade
-------------------------------------------------------------------------------------------------------------
24ADR010     Ananya S             AIDS     24ADT31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADC31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADL31    30       16          53.3% NOT APPROVED BARRED (Grade SA)
24CER003     Aravind K            CIVIL    24CET31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEC31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEL31    30       21          70.0% NOT APPROVED PENDING (Grade SA)
-------------------------------------------------------------------------------------------------------------
 Note: <65% Attendance: Shortage of Attendance (SA - Barred) | 65-74.9%: Requires Medical Condonation
=============================================================================================================


Press [Enter] to return to the Main Menu...
```

STUDENT SEMESTER MARK SHEET:

```
Enter Student Roll Number (e.g., 24ITR001): 24ITR099
Enter Semester (e.g., 3): 3

=================================================================================================================
                                  KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                                      PERUNDURAI, ERODE - 638060
                                OFFICE OF THE CONTROLLER OF EXAMINATIONS
                             GRADE SHEET - REGULATION 2024 (AUTONOMOUS)
=================================================================================================================
 Roll Number : 24ITR099          Student Name : Murasoli B                      Semester : 3
 Department  : IT                Regulation   : R2024 (KEC)                      Date     : 20-10-2026
-----------------------------------------------------------------------------------------------------------------
Course    Course Title               Type      Cr   Attd%    CIA1   CIA2   Asg    Sc.CIA   ESE(100) Sc.ESE  Total  Grd  Result
-----------------------------------------------------------------------------------------------------------------
24ITC31   Full Stack Development      EMBEDDED  4      96.7% 94.0   96.0   95.0      47.50 94.0     47.0     94.50 O    PASS
24ITL31   Programming in Java Labo...  PRACTICAL 1     100.0% 98.0   98.0   100.0     59.16 96.0     38.4     97.56 O    PASS
24ITT32   Programming in Java         THEORY    3      97.8% 90.0   92.0   94.0      36.76 90.0     54.0     90.76 O    PASS
-----------------------------------------------------------------------------------------------------------------
 Total Enrolled Credits : 8      |   Credits Earned : 8      |   Weighted Grade Points : 80.00
 SEMESTER GRADE POINT AVERAGE (SGPA) : 10.00 / 10.00
 EQUIVALENT PERCENTAGE               : 100.00%
 SEMESTER EVALUATION OUTCOME         : [PASSED] ALL REGISTERED COURSES CLEARED
=================================================================================================================
 Grading Standard (R2024): O(91-100)->10 | A+(81-90)->9 | A(71-80)->8 | B+(61-70)->7 | B(50-60)->6 | RA/SA/AB->0
 Rules: Theory (40 CIA/60 ESE, ESE Min 27/60) | Practical (60 CIA/40 ESE, ESE Min 20/40) | Pass Minimum: 50 Marks
=================================================================================================================


Press [Enter] to return to the Main Menu...
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
Enter your choice (1-8): 7

Enter Semester (e.g., 3): 3

=================================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                      DEPARTMENT-WISE CLASS PERFORMANCE & RESULT SUMMARY
                                      SEMESTER: 3
=================================================================================================
Department   Total Students   Passed All     With Arrears   Pass Rate (%)  Class Avg SGPA
-------------------------------------------------------------------------------------------------
AIDS         1                0              1                     0.00%           0.00
CHEM         1                1              0                   100.00%           9.11
CIVIL        1                0              1                     0.00%           8.44
CSE          2                2              0                   100.00%           8.42
ECE          1                1              0                   100.00%           9.00
EEE          1                1              0                   100.00%           9.00
IT           5                5              0                   100.00%           8.47
MECH         1                0              1                     0.00%           6.58
=================================================================================================


Press [Enter] to return to the Main Menu...
```

DATABASE:

STUDENTS DATABASE:

```
mysql> USE kec_eval;
Database changed
mysql> SELECT * FROM students;
+----------+--------------------+------------+------------------+---------------------+
| roll_no  | name               | department | current_semester | created_at          |
+----------+--------------------+------------+------------------+---------------------+
| 22ADR071 | Sudharsan G        | AIDS       |                7 | 2026-10-19 21:29:39 |
| 22ADR081 | Akilesh R          | AIDS       |                8 | 2026-10-19 21:29:39 |
| 22CER071 | Saranya V          | CIVIL      |                7 | 2026-10-19 21:29:39 |
| 22CER081 | Yogeshwaran T      | CIVIL      |                8 | 2026-10-19 21:29:39 |
| 22CHR071 | Swetha T           | CHEM       |                7 | 2026-10-19 21:29:39 |
| 22CHR081 | Balamurugan S      | CHEM       |                8 | 2026-10-19 21:29:39 |
| 22CSR071 | Praveen Kumar R    | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR072 | Priyanga M         | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR081 | Venkatesh S        | CSE        |                8 | 2026-10-19 21:29:39 |
| 22CSR082 | Vidhya R           | CSE        |                8 | 2026-10-19 21:29:39 |
| 22ECR071 | Rajesh K           | ECE        |                7 | 2026-10-19 21:29:39 |
| 22ECR081 | Vigneshwar B       | ECE        |                8 | 2026-10-19 21:29:39 |
| 22EER071 | Rithika S          | EEE        |                7 | 2026-10-19 21:29:39 |
| 22EER081 | Vishnu Vardhan M   | EEE        |                8 | 2026-10-19 21:29:39 |
| 22ITR071 | Manikandan P       | IT         |                7 | 2026-10-19 21:29:39 |
| 22ITR072 | Nandhini S         | IT         |                7 | 2026-10-19 21:29:39 |
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| course_code | title                                         | credits | course_type | semester | created_at          |
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| 24ADC31     | Artificial Intelligence & Knowledge Rep.      |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24ADC41     | Machine Learning Algorithms & Systems         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24ADC51     | Natural Language Processing (NLP)             |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24ADC61     | Generative AI & Large Language Models         |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24ADC71     | Reinforcement Learning & Autonomous Agents    |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24ADL31     | Python for AI & Data Science Laboratory       |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24ADL41     | Machine Learning & Data Mining Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24ADL51     | Deep Learning & NLP Laboratory                |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24ADL61     | Computer Vision & GenAI Laboratory            |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24ADP71     | AI Capstone Project Phase I                   |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
| 24ADP81     | AI Capstone Project Phase II / Industry Track |      10 | PRACTICAL   |        8 | 2026-10-19 20:59:10 |
| 24ADT31     | Mathematical Foundations for Data Science     |       4 | THEORY      |        3 | 2026-10-19 20:59:10 |
| 24ADT41     | Statistical Learning & Data Mining            |       4 | THEORY      |        4 | 2026-10-19 20:59:10 |
| 24ADT51     | Deep Learning & Neural Networks               |       4 | THEORY      |        5 | 2026-10-19 20:59:10 |
| 24ADT61     | Computer Vision & Video Analytics             |       4 | THEORY      |        6 | 2026-10-19 20:59:10 |
| 24ADT71     | AI Ethics, Governance & MLOps Infrastructure  |       3 | THEORY      |        7 | 2026-10-19 20:59:10 |
| 24CEC31     | Surveying and Geomatics                       |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24CEC41     | Fluid Mechanics & Hydraulic Machinery         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24CEC51     | Soil Mechanics & Geotechnical Engineering     |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24CEC61     | Environmental Engineering & Water Treatment   |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24CEC71     | Transportation Engineering & GIS              |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24CEL31     | Surveying Field Work Laboratory               |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24CEL41     | Materials Testing & Hydraulics Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24CEL51     | Geotechnical Engineering Lab                  |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24CEL61     | Environmental Engineering Laboratory          |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24CEP71     | Project Work Phase I                          |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | total_classes | attended_classes | is_condoned | created_at          |
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
|             1 | 24ITR001 | 24ITT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             2 | 24ITR001 | 24ITC31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             3 | 24ITR001 | 24ITT32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             4 | 24ITR001 | 24ITC32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             5 | 24ITR001 | 24ITT33     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|             6 | 24ITR001 | 24MNT31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|             7 | 24ITR001 | 24ITL31     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|             8 | 24ITR001 | 24ITL32     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|             9 | 24ITR001 | 24ITL33     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            10 | 24ITR002 | 24ITT31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            11 | 24ITR002 | 24ITC31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            12 | 24ITR002 | 24ITT32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            13 | 24ITR002 | 24ITC32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            14 | 24ITR002 | 24ITT33     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            15 | 24ITR002 | 24MNT31     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            16 | 24ITR002 | 24ITL31     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|            17 | 24ITR002 | 24ITL32     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            18 | 24ITR002 | 24ITL33     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            19 | 24ITR003 | 24ITT31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            20 | 24ITR003 | 24ITC31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            21 | 24ITR003 | 24ITT32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            22 | 24ITR003 | 24ITC32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            23 | 24ITR003 | 24ITT33     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            24 | 24ITR003 | 24MNT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            25 | 24ITR003 | 24ITL31     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            26 | 24ITR003 | 24ITL32     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            27 | 24ITR003 | 24ITL33     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|            28 | 24CSR001 | 24CST31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            29 | 24CSR001 | 24CSC31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            30 | 24CSR001 | 24CST32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

1. Oracle, Java SE Documentation – JDBC Overview and Architecture. https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/ .

2. Oracle, Java SE API – HttpServer. https://docs.oracle.com/en/java/javase/17/docs/api/jdk.httpserver/com/sun/net/httpserver/HttpServer.html .

3. Oracle, MySQL 8.0 Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/ .

4. Oracle, MySQL Connector/J Developer Guide. https://dev.mysql.com/doc/connector-j/8.0/en/ .

5. University Grants Commission, Guidelines for Choice Based Credit System (CBCS). https://www.ugc.gov.in/pdfnews/8023719_Guidelines-for-CBCS.pdf .

6. Chart.js, Chart.js Documentation. https://www.chartjs.org/docs/latest/ .

7. Tailwind CSS, Tailwind CSS Documentation. https://tailwindcss.com/docs .

8. GeeksforGeeks, Establishing JDBC Connection in Java. https://www.geeksforgeeks.org/establishing-jdbc-connection-in-java/ .

9. YouTube, Java JDBC with MySQL Tutorial Videos. https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

10. Anthropic, Claude AI Assistant (used for project ideas and guidance). https://claude.ai/ .